EXP 17

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clc;
clear;
close all;

% Performance Evaluation of Digital Communication Systems
% Using Eye Diagram Analysis under AWGN

%% 1. System Parameters
Nbits = 10000;          % Number of bits
SPS = 20;               % Samples per symbol
SNR_dB = [0 2 4 6 8 10]; % SNR values in dB
Fs = 10000;             % Sampling frequency (Hz)
Rb = Fs/SPS;            % Bit rate

rng(1);                 % Reproducible random data

%% 2. Generate Random Binary Data
bits = randi([0 1],1,Nbits);

%% 3. BPSK Modulation
% Map bits: 0 -> -1, 1 -> +1
symbols = 2*bits - 1;

% Rectangular pulse shaping
tx = repelem(symbols,SPS);

%% 4. Plot Transmitted Signal
figure('Name','Digital Communication AWGN Analysis');

subplot(3,1,1);
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plot((0:10*SPS-1)/Fs,tx(1:10*SPS),'LineWidth',1.2);
title('Transmitted BPSK Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

%% 5. Add AWGN and Evaluate BER
BER = zeros(size(SNR_dB));
rxStore = cell(size(SNR_dB));

for k = 1:length(SNR_dB)

    % BPSK has one bit per symbol.
    % Convert Eb/N0 to noise variance for unit-energy symbols.
    EbN0 = 10^(SNR_dB(k)/10);
    noiseVariance = 1/(2*EbN0);

    noise = sqrt(noiseVariance)*randn(size(tx));
    rx = tx + noise;
    rxStore{k} = rx;

    % Sample at symbol centers
    sampleIndex = floor(SPS/2)+1:SPS:length(rx);
    receivedSymbols = rx(sampleIndex);

    % BPSK detection
    detectedBits = receivedSymbols > 0;

    % Calculate BER
    BER(k) = mean(detectedBits ~= bits);
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end

%% 6. Eye Diagram Before Noise
figure('Name','Eye Diagram Analysis');

subplot(2,1,1);
eyeSamples = tx(1:2000);
plot(reshape(eyeSamples,SPS*2,[]),'b');
title('Eye Diagram Without Noise');
xlabel('Sample Index');
ylabel('Amplitude');
grid on;

%% 7. Eye Diagram Under AWGN
% Use an intermediate SNR value
selectedIndex = 3;  % 4 dB
rxEye = rxStore{selectedIndex};

subplot(2,1,2);
eyeLength = 2*SPS;
numberOfTraces = 100;

hold on;
for n = 1:numberOfTraces
   startIndex = (n-1)*SPS+1;
   if startIndex+eyeLength-1 <= length(rxEye)
      segment = rxEye(startIndex:startIndex+eyeLength-1);
      plot((0:eyeLength-1)/SPS,segment,'b');
   end
end
title(['Eye Diagram with AWGN at SNR = ', ...
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    num2str(SNR_dB(selectedIndex)),' dB']);
xlabel('Time (symbol periods)');
ylabel('Amplitude');
grid on;
hold off;

%% 8. Plot BER Performance
figure('Name','BER Performance');

semilogy(SNR_dB,BER,'ro-','LineWidth',1.5);
hold on;

% Theoretical BER for coherent BPSK in AWGN
theoreticalBER = 0.5*erfc(sqrt(10.^(SNR_dB/10)));
semilogy(SNR_dB,theoreticalBER,'b--','LineWidth',1.5);

title('BER Performance of BPSK under AWGN');
xlabel('Eb/N0 (dB)');
ylabel('Bit Error Rate (BER)');
legend('Simulated BER','Theoretical BER');
grid on;
ylim([1e-5 1]);

%% 9. Display Results
fprintf('\nDigital Communication System Performance\n');
fprintf('----------------------------------------\n');
fprintf('Eb/N0 (dB)      Simulated BER\n');
fprintf('----------------------------------------\n');

for k = 1:length(SNR_dB)
    fprintf('%6.1f         %.6f\n',SNR_dB(k),BER(k));
```

end

disp('Eye diagram and AWGN performance analysis completed.');

